

Course Code: B23CS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DEEP LEARNING					
(For CSE)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Deep Learning.	1	1	2
	b).	What is model evaluation?	1	1	2
	c).	What is an Artificial Neuron?	2	1	2
	d).	Define optimization in deep learning.	2	1	2
	e).	What is Theano?	3	1	2
	f).	What is deep learning workstation?	3	1	2
	g).	Define multichannel convolution.	4	1	2
	h).	List any two applications of CNN.	4	1	2
	i).	What is Deep Reinforcement Learning?	5	1	2
	j).	Define Deep Belief Network	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain Random Forests and their advantages.	1	2	5
	b).	Explain the types of machine learning with examples.	1	2	5
OR					
3.	a).	Explain probabilistic modeling in machine learning.	1	2	5
	b).	Explain overfitting and underfitting with suitable examples.	1	2	5
UNIT-2					
4.	a).	Explain the structure of a biological neuron and compare it with an artificial neuron model.	2	2	5
	b).	Explain different activation functions used in neural networks.	2	2	5
OR					
5.	a).	Describe the architecture and components of an Artificial Neural Network (ANN).	2	2	5
	b).	Discuss optimization techniques used in deep learning models.	2	2	5
UNIT-3					
6.	a).	Demonstrate how TensorFlow, Theano, and CNTK can be used to implement deep learning models.	3	3	5
	b).	Apply neural networks to perform binary classification for sentiment	3	3	5

		analysis of movie reviews.			
		OR			
7.	a).	Illustrate the anatomy of a neural network and apply it to model a simple learning task.	3	3	5
	b).	Implement a neural network model for multiclass classification of newswire data.	3	3	5
		UNIT-4			
8.	a).	Explain the architecture and working of Recurrent Neural Networks (RNN).	4	2	5
	b).	Describe the implementation of CNN using PyTorch.	4	2	5
		OR			
9.	a).	Describe the architecture of Convolutional Neural Networks (CNN).	4	2	5
	b).	Discuss the advantages and limitations of RNNs.	4	2	5
		UNIT-5			
10.	a).	Explain the role of deep learning in computer vision applications.	5	2	5
	b).	Describe the concept of Deep Reinforcement Learning.	5	2	5
		OR			
11.	a).	Explain the architecture of Generative Adversarial Networks (GANs)	5	2	5
	b).	Explain Boltzmann Machines and Restricted Boltzmann Machines.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Estd. 1980

AUTONOMOUS

Course Code: B23HS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HUMAN RESOURCE & PROJECT MANAGEMENT					
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the role of HR manager.	1	1	2
	b).	Define recruitment.	1	1	2
	c).	List any two models of HR accounting.	2	1	2
	d).	Define career counseling.	2	1	2
	e).	Define Resource management.	3	1	2
	f).	State the 80-20 rule.	3	1	2
	g).	Define Brainstorming.	4	1	2
	h).	Write about Cash flow analysis.	4	1	2
	i).	List any two forms of project organization.	5	1	2
	j).	Define project control.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Interpret the nature and scope of Human Resource Management.	1	2	5
	b).	Discuss the emerging trends in HRM and their impact on organizations.	1	2	5
OR					
3.	a).	Explain the ethical aspects of HRM with suitable examples.	1	2	5
	b).	Describe the steps involved in the selection procedure.	1	2	5
UNIT-2					
4.	a).	Discuss the different methods of training used in organizations.	2	2	5
	b).	Explain the importance of performance appraisal in organizations.	2	2	5
OR					
5.	a).	Summarize the concept of career development and its importance.	2	2	5
	b).	Explain group interaction and its importance in HRD.	2	2	5
UNIT-3					
6.	a).	Explain the concept of Project Management and its importance.	3	2	5
	b).	Discuss the different types of projects with examples.	3	2	5
OR					

7.	a).	Explain the stages of the project life cycle.	3	2	5
	b).	Interpret the process of project selection.	3	2	5
		UNIT-4			
8.	a).	Describe the criteria used for project selection.	4	2	5
	b).	A company invests ₹2,00,000 in a project. The expected cash inflows are: Year 1: ₹50,000 Year 2: ₹60,000 Year 3: ₹70,000 Year 4: ₹80,000 Calculate the Payback Period.	4	2	5
		OR			
9.		A project requires ₹2,00,000 investment. It generates ₹80,000 annually for 4 years. Discount rate = 10% Calculate the NPV and state whether the project is acceptable.	4	2	10
		UNIT-5			
10.	a).	Interpret the process of project planning in detail.	5	2	5
	b).	Explain the human aspects of project management.	5	2	5
		OR			
11.	a).	Discuss the prerequisites for successful project implementation.	5	2	5
	b).	Summarize the importance of project review.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CS4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE ARCHITECTURE & DESIGN PATTERNS					
(Common to CSE,AIDS, CSBS, CSD)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the role of Model–View–Controller (MVC) in design patterns.	1	2	2
	b).	Differentiate between cohesion and coupling.	1	2	2
	c).	Explain how actors are identified in use-case analysis.	2	2	2
	d).	Explain the concept of conceptual classes in system analysis.	2	2	2
	e).	Summarize the key idea behind the Adapter pattern.	3	2	2
	f).	Describe how the Proxy pattern controls access to another object.	3	2	2
	g).	Explain the role of the Model component in MVC architecture.	4	2	2
	h).	Explain how the undo operation can be supported in an interactive system.	4	2	2
	i).	Explain the concept of a distributed object system.	5	2	2
	j).	Explain the roles of client and server in a distributed application.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the essential elements used to document a design pattern.	1	2	5
	b).	Describe how design patterns help solve common design problems in software development.	1	2	5
		OR			
3.	a).	Describe the key concepts of object-oriented design.	1	2	5
	b).	Describe the drawbacks of the object-oriented paradigm.	1	2	5
		UNIT-2			
4.	a).	Use requirement gathering techniques to prepare a functional requirement specification for a student attendance management system.	2	3	5
	b).	Demonstrate how use-case analysis can be used to represent the functional requirements of a bank ATM system.	2	3	5
		OR			
5.	a).	Illustrate the process of identifying conceptual classes and relationships from a given problem description.	2	3	5
	b).	Apply domain knowledge to refine the analysis model of an e-	2	3	5

		commerce order processing system.			
		UNIT-3			
6.	a).	Demonstrate how the Bridge pattern can be used to separate device abstraction from its implementation in a remote-control system.	3	3	5
	b).	Illustrate the use of the Decorator pattern to dynamically add additional features to a text editor component.	3	3	5
		OR			
7.	a).	Apply the Facade pattern to simplify the interaction between multiple subsystems in a banking application.	3	3	5
	b).	Demonstrate how the Flyweight pattern can be used to optimize memory usage in a large document editor with many repeated characters.	3	3	5
		UNIT-4			
8.	a).	Apply the MVC architectural pattern to design an interactive drawing application, identifying the model, view, and controller components.	4	3	5
	b).	Apply MVC principles to design the subsystems of an interactive online shopping interface.	4	3	5
		OR			
9.	a).	Demonstrate how a new feature can be added to an MVC-based system without modifying the entire architecture.	4	3	5
	b).	Illustrate how pattern-based solutions improve extensibility in an MVC-based interactive system.	4		5
		UNIT-5			
10.	a).	Demonstrate how Java Remote Method Invocation (RMI) can be used to enable communication between distributed objects in a networked application.	5	3	5
	b).	Demonstrate how RESTful web services can be used to provide data access in a student information system.	5	3	5
		OR			
11.	a).	Illustrate the process of implementing an object-oriented system on the web for an online shopping application.	5	3	5
	b).	Illustrate how an Enterprise Service Bus (ESB) can be used to integrate different enterprise applications in a distributed environment.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CS4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
BLOCKCHAIN TECHNOLOGY					
(Common to CSE, AIML)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the origin of blockchain.	1	2	2
	b).	Explain different types of blockchain.	1	2	2
	c).	Explain public blockchain.	2	2	2
	d).	Explain different types of Oracles	2	2	2
	e).	Explain Byzantine Fault.	3	2	2
	f).	Discuss the advantages of Ripple.	3	2	2
	g).	Explain Security Aspects in Bitcoin	4	2	2
	h).	Explain Limitations of Blockchain	4	2	2
	i).	Discuss How to Use Python Online Editor	5	2	2
	j).	Explain the Chain codes.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the Components of Blockchain.	1	2	5
	b).	Identify the importance of Consensus Protocol in the Blockchain.	1	2	5
		OR			
3.	a).	Explain Decentralization and Distribution in the Blockchain.	1	2	5
	b).	Explain different types of Cryptocurrencies.	1	2	5
		UNIT-2			
4.	a).	Explain Public Blockchain system.	2	2	5
	b).	Discuss different Characteristics of a Smart Contract	2	2	5
		OR			
5.	a).	Explain the Ethereum Blockchain system.	2	2	5
	b).	Explain Smart Contracts in Ethereum.	2	2	5
		UNIT-3	3	2	5
6.	a).	Explain Key Characteristics of Private Blockchain.	3	2	5
	b).	Discuss different Blockchain Fundraising Methods.			
		OR			

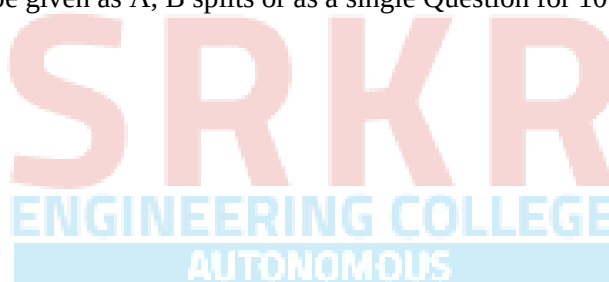
7.	a).	Explain Characteristics of Consortium Blockchain	3	2	5
	b).	Discuss Pros and Cons of Initial Coin Offering	3	2	5
		UNIT-4			
8.	a).	Explain the Security and Privacy Challenges of Blockchain.	4	2	5
	b).	Illustrate how Blockchain can be used in Banking and Finance.	4	3	5
		OR			
9.	a).	Discuss the process of Safeguarding Blockchain Smart Contract.	4	2	5
	b).	Illustrate how Blockchain can be used in Internet of Things	4	3	5
		UNIT-5			
10.	a).	Illustrate how Python Packages can be used for Blockchain.	5	3	5
	b).	Illustrate Blockchain Application development Using Fabric Java SDK.	5	3	5
		OR			
11.	a).	Explain different Components of Hyperledger Fabric Network	5	2	5
	b).	Illustrate how Blockchain can be used in Health Care.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23CS4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AUGUMENTED REALITY & VIRTUAL REALITY					
(Common to CSE, CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Build the history of augmented reality explain in briefly.	1	3	2
	b).	Apply characteristics of tracking technology.	1	3	2
	c).	Create a quick explanation of Outdoor Tracking.	2	3	2
	d).	Select the Scene Graphs and explain them briefly.	2	3	2
	e).	Make the use of human physiology and perception in VR.	3	3	2
	f).	Use visual distortions in virtual reality (VR) temporarily.	3	3	2
	g).	Select the consequences for virtual reality (VR) briefly.	4	3	2
	h).	Choose how to repair optical distortions briefly.	4	3	2
	i).	Select The Vestibular System: A Brief Explanation.	5	3	2
	j).	Choose the Auditory Rendering and briefly explain it.	5	3	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Make use of augmented reality (AR), provide instances of AR applications, and talk about adjacent topics like virtual reality (VR) and mixed reality.	1	3	5
	b).	Apply visual perception to augmented reality (AR) and quickly define multimodal displays.	1	3	5
OR					
3.	a).	Chose the requirements and characteristics of AR displays, as well as the spatial display model and visual displays.	1	3	5
	b).	Make the use of coordinate systems used in AR tracking, as well as the differences between stationary tracking systems and mobile sensors.	1	3	5
UNIT-2					
4.	a).	Choose "How does AR marker track work?" Describe briefly the benefits of tracking natural features.	2	3	5
	b).	Make advantage of the various AR interaction modalities, and briefly discuss the limitations in outside AR tracking.	2	3	5
OR					

5.	a).	Choose one of the following interaction modalities: haptic interaction in augmented reality, virtual user interfaces on actual surfaces and augmented paper, output, input, and tangible interfaces, and explain it succinctly.	2	3	5
	b).	Make use of the scene graph function in augmented reality apps and explain it briefly.	2	3	5
		UNIT-3			
6.	a).	Choose virtual reality (VR) and provide a brief overview of human physiology and perception in virtual reality.	3	3	5
	b).	It explains how to use axis-angle rotations, geometric models, VR viewing transformations, and light and optics, such as lenses, aberrations, the human eye, cameras and displays.	3	3	5
		OR			
7.	a).	Choose the virtual reality (VR) and its main features. How does human perception affect VR design?	3	3	5
	b).	Make use of lenses and optical imperfections. VR. How do cameras and screens work in VR systems?	3	3	5
		UNIT-4			
8.	a).	Identify the routes that lead from photoreceptors to the cortex of the eye. How do the cornea and photoreceptors of the human eye process light?	4	3	5
	b).	Create eye movements and investigate how they influence VR design, depth, motion, and color perception.	4	3	5
		OR			
9.	a).	Develop the visual cortex's role in perception. How do photoreceptors, such as rods and cones, function? How does virtual reality utilize depth perception?	4	3	5
	b).	Select visual rendering techniques such as ray tracing, shading, rasterization, fixing optical aberrations and latency in virtual reality, and creating immersive images and films for VR.	4	3	5
		UNIT-5			
10.	a).	Make use of the visual pathways that connect photoreceptors and the visual cortex. How does light pass from the cornea to the photoreceptors?	5	3	5
	b).	Use eye movements, types, and implications for VR design, and in-depth perception of VR cues (binocular disparity, etc.) to describe it.	5	3	5
		OR			
11.	a).	Make use of the way motion is perceived. How VR uses motion perception, color perception, the influence of color vision in VR, and rendering in VR: ray tracing versus rasterization to designate it.	5	3	5

	b).	Select the distortion correction option. How and why is it implemented in virtual reality? Outline latency reduction strategies and methods for improving virtual reality.	5	3	5
--	------------	--	----------	----------	----------

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23CS4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTERNET OF THINGS					
(For CSE)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Internet of Things.	1	2	2
	b).	Write the evolution of IoT.	1	2	2
	c).	List any two characteristics of sensors.	2	2	2
	d).	Why is processing important in IoT.	2	2	2
	e).	What is the role of Bluetooth in IoT	3	2	2
	f).	What is IEEE 802.15.4.	3	2	2
	g).	Define interoperability in IoT.	4	2	2
	h).	State any two applications of fog computing.	4	2	2
	i).	List any two challenges associated with IoT.	5	2	2
	j).	What is security challenge in IoT?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the concept and evolution of Internet of Things.	1	2	5
	b).	Explain Machine-to-Machine communication and Cyber Physical Systems in IoT.	1	2	5
		OR			
3.	a).	Explain the role of Wireless Sensor Networks in IoT.	1	2	5
	b).	Explain IoT networking components and addressing strategies	1	2	5
		UNIT-2			
4.	a).	Explain sensors and their characteristics in IoT.	2	2	5
	b).	Explain the importance of processing in IoT.	2	2	5
		OR			
5.	a).	Explain actuators, actuator types, and characteristics.	2	2	5
	b).	Explain different types of sensing and sensing considerations.	2	2	5
		UNIT-3			
6.	a).	Classify various IoT connectivity technologies and explain in detail.	3	3	5
	b).	Utilize the role of RFID, NFC, Wi-Fi, and Bluetooth in IoT.	3	3	5

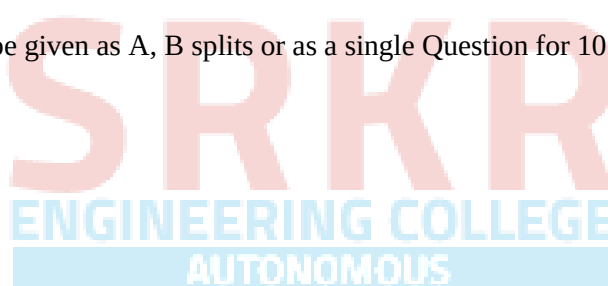
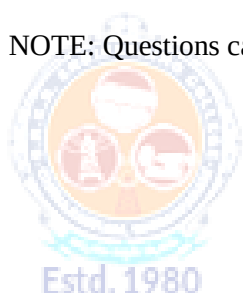
		OR			
7.	a).	Describe IoT communication technologies and their types.	3	3	5
	b).	Describe the working of Zigbee and explain in detail.	3	3	5
		UNIT-4			
8.	a).	Describe IoT standards and frameworks.	4	3	5
	b).	Compare cloud computing and fog computing in IoT.	4	3	5
		OR			
9.	a).	Discuss interoperability in IoT and its importance.	4	3	5
	b).	Describe the role of fog computing in IoT applications.	4	3	5
		UNIT-5			
10.	a).	Outline the emerging pillars and future of IoT.	5	3	5
	b).	Describe Agricultural IoT with suitable examples.	5	3	5
		OR			
11.	a).	Discuss the major challenges associated with IoT.	5	3	5
	b).	Discuss Vehicular IoT and its applications.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23CS4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AGILE METHODOLOGIES					
(Common to CSE,AIIML, AIDS, CSBS, IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	What is meant by Agile manifesto	1	1	2
	b)	Define task board	1	1	2
	c)	Define self organizing team	2	1	2
	d)	Suggest appropriate agile communication principles for a software development team that is struggling with communication gaps and delayed deliveries in Ebook Reader project.	2	3	2
	e)	Mention the three main roles on a Scrum project.	3	1	2
	f)	How is Scrum both incremental and iterative?	3	2	2
	g)	Define pair programming	4	1	2
	h)	What is feedback loops and why do XP teams create them	4	2	2
	i)	Define Shuhari	5	1	2
	j)	What are the types of waste that create constraints in the workflow?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a)	Explain the core values of Agile manifesto and describe how they differ from traditional software development approaches	1	2	5
	b)	Analyze how a “fractured perspective” leads to project failure. How does Agile thinking help resolve this issue?	1	3	5
		OR			
3.	a)	Explain the idea of “Principles over Practices” in Agile. Why is focusing only on practices without understanding underlying principles often ineffective	1	2	5
	b)	Illustrate with an example how “Responding to Change over Following a Plan” benefits a software project.	1	3	5
		UNIT-2			
4.		Explain Agile principles and their importance in software development.	2	2	10
		OR			
5.	a)	Explain how effective communication contributes to better project	2	2	5

		delivery in software development.			
	b)	Explain how continuous improvement contributes to better project and team performance in Agile	2	2	5
		UNIT-3			
6.	a)	Explain the Scrum framework. Discuss the roles of Product Owner, Scrum master, and development team	3	2	5
	b)	Design a complete sprint planning process for a mobile application development project. Apply user stories, story points. Velocity estimation and task board management to create a comprehensive sprint plan.	3	3	5
		OR			
7.	a)	Explain the significance of Scrum values. Explain why Scrum practices alone are insufficient without adherence to these values.	3	2	5
	b)	A sprint fails because the Product Owner frequently changes priorities during sprint. Use Scrum planning rules to analyze the product owner support for sprint success without disrupting execution.	3	3	5
		UNIT-4			
8.		Explain the XP values and their role in shaping an effective development mindset	4	2	10
		OR			
9.	a)	Illustrate how continuous integration helps in maintaining software quality in XP	4	2	5
	b)	Explain how XP practices help teams embrace changing requirements effectively.	4	2	5
		UNIT-5			
10.	a)	Explain Lean thinking and identify different types of waste in software development	5	2	5
	b)	Analyse how limiting Work in Progress (WIP) improves flow in a Kanban system	5	3	5
		OR			
11.	a)	Explain Kanban principles to manage workflow in a software maintenance project	5	2	5
	b)	Evaluate the role of Agile Coach in helping organizations adopt agile practices successfully	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CS4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Generative AI					
(For CSE)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the key difference between Generative Models and Discriminative Models with an example.	1	3	2
	b).	Classify different types of Generative Models and mention one use case for each	1	3	2
	c).	Explain the role of attention mechanisms in Transformer architecture	2	2	2
	d).	Summarize the key differences between encoder-based models (e.g., BERT) and decoder-based models (e.g., GPT)	2	2	2
	e).	Explain the difference between GPT-4V and a text-only language model?	3	2	2
	f).	A user gives a text prompt to generate an image. Which model is more suitable: DALL·E 3 or VAE? Justify your answer.	3	3	2
	g).	A user wants to generate melody notes one after another. Determine which model is more suitable: RNN or CycleGAN? Justify your answer	4	3	2
	h).	Compare polyphonic music generation with simple melody generation?	4	2	2
	i).	Explain how Llama is useful in text generation	5	2	2
	j).	Describe any two uses of Google Copilot	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Apply the concept of Generative Adversarial Networks (GANs) in healthcare and Explain its working	1	3	5
	b).	Demonstrate how Variational Autoencoders (VAEs) can be used and Illustrate the generative process involved in it.	1	3	5
OR					
3.	a).	Apply autoregressive models to a text generation task. Explain how the probabilistic modelling helps in generating coherent sequences.	1	3	5
	b).	Develop a use-case scenario where diffusion models (e.g., Vector Quantized Diffusion Models) can be applied. Explain the generative process and challenges involved	1	3	5

		UNIT-2			
4.	a).	Apply the Transformer architecture to a text generation task. Explain how the encoder, decoder, and attention mechanisms work together to generate meaningful text.	2	3	5
	b).	Demonstrate how prompt engineering techniques can be used to improve responses from large language models. Provide examples and explain the role of Reinforcement Learning from Human Feedback (RLHF).	2	3	5
		OR			
5.	a).	Apply Retrieval-Augmented Generation (RAG) to design a system for Document question answering. Explain the workflow.	2	3	5
	b).	Demonstrate a use-case scenario where a multimodal LLM can be applied and discuss challenges such as hallucination.	2	3	5
		UNIT-3			
6.	a).	Explain how generator and discriminator compete for generating images during the adversarial training process in GANs.	3	2	5
	b).	A user wants to generate an image from the prompt: "A futuristic classroom in watercolor painting style." Determine whether Stable Diffusion or DALL·E 3 is more appropriate for this task, and justify your answer	3	3	5
		OR			
7.	a).	Describe the concept of Visual Transformers (ViT). How are images processed differently in ViT compared to CNN-based models?	3	2	5
	b).	Suppose a GAN model is repeatedly generating very similar images for different prompts. Using your knowledge of GANs, determine the issue and explain how you would address it.	3	3	5
		UNIT-4			
8.	a).	Explain the role of Recurrent Neural Networks (RNNs) in music generation. Why are they suitable for sequence modeling?	4	2	5
	b).	Analyze the advantages and limitations of CycleGAN in painting generation tasks	4	4	5
		OR			
9.	a).	Describe the architecture and purpose of MuseGAN in polyphonic music generation.	4	2	10
		UNIT-5			
10.	a).	Explain the purpose of Hugging Face AutoTrain in model training.	5	2	5
	b).	A user wants to generate text using an open-source model. Explain how Llama can be used for this purpose.	5	3	5
		OR			

11.	a).	Explain the role of LangChain in Generative AI applications	5	2	5
	b).	A student wants to run a Generative AI model on a laptop without internet. Determine whether GPT4All can be used or not, justify it.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23CS4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER VISION					
(Common to CSE, CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is Radiometry? Why is it important in computer vision?	1	2	2
	b).	Define surface color.	1	1	2
	c).	What is the Fourier Transform?	2	1	2
	d).	What is meant by edge detection?	2	1	2
	e).	Define stereopsis.	3	1	2
	f).	What are Gestalt principles?	3	2	2
	g).	How are lines fitted using the Hough Transform?	4	2	2
	h).	What is Kalman Filtering?	4	1	2
	i).	What are camera parameters?	5	1	2
	j).	What is pose clustering?	5	1	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Explain the Pinhole Camera Model with a neat diagram and derive the projection equation.	1	2	5
	b).	Explain Photometric Stereo and its applications.	1	2	5
		OR			
3.	a).	Discuss Human Color Perception and the role of cones and rods	1	2	5
	b).	Compare Local Shading Models and Global Shading Models.	1	2	5
		UNIT-II			
4.	a).	Explain how filters are used as templates for pattern detection.	2	2	5
	b).	Discuss analysis and synthesis using oriented pyramids.	2	2	5
		OR			
5.	a).	Explain Spatial Frequency and Fourier Transforms in image processing.	2	2	5
	b).	Describe Shape from Texture and its applications.	2	2	5
		UNIT-III			

6.	a).	What is image segmentation? Explain its importance in image processing.	3	2	5
	b).	Explain different applications of segmentation in computer vision.	3	2	5
		OR			
7.	a).	Discuss the Gestalt principles used in human vision for grouping.	3	2	5
	b).	Discuss segmentation using graph-theoretic clustering methods.	3	2	5
		UNIT-IV			
8.	a).	Explain the concept of robustness in model fitting.	4	2	5
	b).	Explain Linear Dynamic Models used in object tracking.	4	2	5
		OR			
9.	a).	Describe the missing data problem in segmentation and fitting.	4	2	5
	b).	Discuss the Kalman Filtering algorithm with its applications.	4	2	5
		UNIT-V			
10.	a).	Explain the linear approach to camera calibration.	5	2	5
	b).	Explain analytical photogrammetry and its applications.	5	2	5
		OR			
11.	a).	Explain the linear approach to camera calibration.	5	2	5
	b).	Discuss the Least Squares Parameter Estimation method used in calibration.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

IV B.Tech. I Semester MODEL QUESTION PAPER

CYBER PHYSICAL SYSTEMS

(Common to IT, CSE, CSD)

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE** Question from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define Cyber Physical system	1	1	2
	b).	What are the Basic Techniques used in Cyber physical systems.	1	1	2
	c).	Define attack models in CPS security	2	1	2
	d).	Justify security importance in cyber-physical systems?	2	2	2
	e).	What is synchronization in distributed CPS?	3	1	2
	f).	Discuss fault tolerance challenges in distributed CPS	3	2	2
	g).	Differentiate between hard real-time and soft real-time systems	4	2	2
	h).	Explain the role of deadlines in scheduling	4	2	2
	i).	What is model integration in CPS?	5	1	2
	j).	Discuss multidisciplinary interactions in CPS design	5	2	2

5 x 10 = 50 Marks

		UNIT-1			
2.	a).	Differentiate between traditional control systems and cyber-physical systems	1	2	5
	b).	Explain the motivation behind symbolic control design	1	2	5
		OR			
3.	a).	What are the challenges in controller synthesis for CPS?	1	1	5
	b).	Discuss safety and reachability specifications in CPS	1	2	5
		UNIT-2			
4.	a)	Explain the security challenges of Cyber-Physical Systems.	2	2	5
	b)	What are the major threats to CPS? Explain	2	1	5
		OR			
5.	a).	Explain redundancy as a countermeasure in CPS	2	2	5
	b).	Differentiate between insider and outsider attacks	2	2	5
		UNIT-3			

6.	a).	Discuss reliability and consistency issues in distributed CPS	3	2	5
	b).	Explain event-based synchronization methods	3	1	5
		OR			
7.	a).	Explain the role of formal software engineering in CPS	3	2	5
	b).	Explain the working of consensus algorithms	3	2	5
		UNIT-4			
8.	a).	Define Real-Time Systems and explain their importance in CPS	4	2	5
	b).	Explain fixed timing parameters in real-time scheduling	4	2	5
		OR			
9.	a).	Explain the role of memory in real-time scheduling	4	2	5
	b).	Explain variability and uncertainty in CPS scheduling	4	2	5
		UNIT-5			
10.	a).	Explain the motivation behind integrated modeling approaches	5	2	5
	b).	Discuss components and connectors in CyPhyML	5	2	5
		OR			
11.	a).	Discuss synchronous and asynchronous timing models.	5	2	5
	b).	Explain causal relationships in CPS models	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

